

APPENDIX A. KRIPKE SEMANTICS AND NESTED MODALITIES

The modal semantics can be construed as an example of Kripke semantics. We define it as follows:

Definition A.1. Let Θ be a topological space, c a confidence function, and Ω a sample space. Evidence E is an element $E \in \Omega^{<\omega}$.

A possible world is specified by a pair of $\theta_w \in \Theta$ and $E_w \in \Omega^{<\omega}$: $w = (\theta_w, E_w)$.

The “ c -plausibility” accessibility relation between two worlds is given by

$$(24) \quad w R_c v \Leftrightarrow \theta_v \in c(E_w)$$

The “extendibility” accessibility relation is given by

$$(25) \quad w R_E v \Leftrightarrow E_w \subseteq E_v$$

Let $H \in 2^\Theta$. We say that $v \Vdash H$ just in case $\theta_v \in H$.

We represent the corresponding modal operators of these accessibility relations by $\Box_c$ and $\Box_E$ respectively.

It is straightforward to verify that the Kripke semantics of $\Box_c$ agrees with the semantics of unnested modal sentences given in Section 3.2.

Proposition A.2. Let $H \subseteq \Omega$. Then for every w , $w \Vdash \Box_c H$ if and only if $c(E_w) \subseteq H$.

Proof. By definition, $w \Vdash \Box_c H$ if and only if, for all v such that $w R_c v$, $v \Vdash H$. By definition of R_c , if $w R_c v$ then $\theta_v \in c(E_w)$. Therefore $w \Vdash \Box_c H$ if and only if $\theta_v \in H$ for all v accessible by w , if and only if $c(E_w) \subseteq H$. $\square$

Whereas nested modalities were left undefined in Section 3.2, this frame semantics yields truth conditions for nested modal formulas.

Under mild conditions, we can make sense of a key aspect of statistical epistemology:

Definition A.3. We say that a confidence function c satisfied the **precise extension property (PEP)** provided that for each $\theta \in \Theta$ and $U \in \mathcal{O}(\Theta)$ containing θ and $E \in \Omega^{<\omega}$ there exists an $E' \supseteq E$ such that $\theta \in c(E') \subseteq U$.

Wald confidence intervals satisfy the PEP. Under the assumption of the PEP, we can precisely formulate epistemic limits to statistical confirmation. In other words, we can prove a formal expression of the problem of induction specific to hypothesis testing.

Proposition A.4. Let $H \subseteq \Theta$ be nonempty. Suppose further that c satisfies the PEP. Then for every w

$$(26) \quad w \Vdash \Diamond_E \Diamond_c H.$$

In particular, if H and H^c are both nonempty, then for every w

$$w \Vdash (\Diamond_E \Diamond_c H) \wedge (\Diamond_E \Diamond_c H^c).$$

Proof. Suppose $w = (\theta_w, E_w)$ is a possible world. Suppose $\theta_0 \in H$. By the PEP, for every $\theta_v \in c(E_w)$ there is some extension $E' \supseteq E_w$ such that $\theta_0 \in c(E')$. Let $v = (\theta_0, E')$. Then $w R_c v$, realizing $\Diamond_c H$. Thus

$$w \Vdash \Diamond_E \Diamond_c H.$$

$\square$

That is, it is always possible to extend evidence to alter our position from $\Box_c H$ to $\Diamond_c \neg H$.

APPENDIX B. CONFIRMATION OF POINT HYPOTHESES

The confirmability of hypotheses through frequentist methods is a problem that has drawn considerable interest. Prior work on the characterization of confirmable or verifiable hypotheses has been done by Genin.

In the context of Two One-Sided Testing, a form of Equivalence Testing, Lakens' makes the claim that "it is statistically impossible to support the hypothesis that a true effect size is exactly zero." [5].

One possible argument in favor of this position is that the hypothesis $\theta = 0$ is *sharp*: for any given piece of data sufficiently compatible with $\theta = 0$, there exist θ' near θ also compatible with the data.

One might argue this using standard confidence intervals. For example, when comparing the difference in proportions between two subgroups A and B , one might be tempted to say that evidence in favor of H is given just in case the complement H^c is rejected. Under the duality between hypothesis testing and confidence interval estimation, a hypothesis J is rejected in case event E is witnessed just in case $J \cap c(E) = \emptyset$. If one is to equate confirmation of H with rejection of the complement H^c , H is confirmed just in case $c(E) \subseteq H$.

The typical confidence intervals employed in practice are either open subsets of the space of model parameters (e.g. Wald confidence intervals $(l(E), u(E))$) or have nonempty interior (e.g. bootstrap confidence intervals $[l(E), u(E))$). In this context, the irrefutability of the negation of a point hypothesis is immediate on topological grounds: a point hypothesis $H = \{\theta_0\}$ has empty interior while confidence intervals have nonempty interiors. Since having nonempty interior is preserved under supersets, point sets are not confirmable under this inferential schema.

However, this discussion relies on a crucial premise: why assume in the first place that a confidence region necessarily have open interior? More broadly speaking, we investigate how sensitive to the choice of a confidence region process c the question of confirmation is.

B.1. Rigged Confidence Regions. We now exhibit a construction of confidence region processes that have the possibility of confirming *arbitrary* hypotheses while retaining both the confidence property and the stronger topological confidence property.

Let Θ be a set of probability measures on a sigma algebra Σ on Ω topologized with setwise convergence. Suppose that $c : \Sigma \rightarrow 2^\Theta$ satisfies the confidence property at level $\alpha/2$ and that each $c(E)$ is an open subset of Θ . Thus c satisfies the topological confidence property at level $\alpha/2$.

Let $H \subset 2^\Theta$ be an arbitrary hypothesis and E_0 an event in Σ . We define the (E_0, H_0) -rigged confidence region c^* as follows:

$$c^*(E) = \begin{cases} c(E) & \text{if } E \neq E_0 \\ H & \text{if } E = E_0. \end{cases}$$

Suppose that there exists an event E_0 such that

$$0 < \limsup_{\theta \in \Theta} (\mathbb{P}_\theta(E_0)) = \gamma < \frac{\alpha}{2}.$$

Then the function c^* satisfies the α topological confidence property, since for all θ

$$\begin{aligned} \mathbb{P}_\theta(\theta \notin c(E) \text{ or } c(E) \text{ has empty interior}) &\leq \mathbb{P}_\theta(\theta \notin c(E)) + \mathbb{P}_\theta(c(E) \text{ has empty interior}) \\ (27) \quad &\leq \frac{\alpha}{2} + \gamma \\ &\leq \alpha \end{aligned}$$

Example B.1. Let $\Omega = 2^7$ and $\Sigma = 2^{2^7}$. Let $\Theta = [0, 1]$ and interpret $\mathbb{P}_\theta(E_s) = \theta^{|s|_1}(1 - \theta)^{|s|_0}$. Let $\alpha = 0.05$ and let c be the Wald confidence interval at level $\alpha' = 0.04$. Consider the event $E_{1011000}$. Then

$$\begin{aligned} \limsup_{\theta \in \Theta} (\mathbb{P}_\theta(E_{1011000})) &= \left(\frac{3}{7}\right)^3 \left(\frac{4}{7}\right)^4 \\ (28) \quad &\approx 0.0084 \end{aligned}$$

since $\theta = \frac{3}{7}$ maximizes the probability of the event $E_{1011000}$.

Then the rigged confidence region c^* satisfies the topological confidence property at level $\alpha^* \leq 0.0485$ by Inclusion-Exclusion. $\square$

In other words, one can rig an open-valued confidence region of level $\alpha/2$ to confirm

B.2. The Topology of the Confidence Property. Let Θ be a collection of measures on a σ -algebra Σ on Ω , and let $c : \Sigma \rightarrow 2^\Theta$ be a *confidence region* function. For example, if $\Omega = 2^\omega$ and Σ is the σ -algebra generated by basic open sets in the discrete topology.

Definition B.2. We say that c has the confidence property at level α provided that every for every E the event $\theta \in c(E)$ is in Σ and

$$\inf_{\theta \in \Theta} \mathbb{P}_\theta(\theta \in c(E)) \geq 1 - \alpha.$$

$\square$

In case Θ has topological structure, the confidence property imposes constraints on the topological structure of the confidence region. We assume that Θ is topologized by setwise(or “strong”) convergence of measures; that is,

$$\theta_n \rightarrow_{sw} \theta \Leftrightarrow \mathbb{P}_{\theta_n}(A) \rightarrow \mathbb{P}_\theta(A) \text{ for all } A \in \Sigma^{\otimes n}.$$

In other words, setwise convergence holds just in case the limit of probabilities of each event A converge to the probability of the limit.

This is a natural topology for statistical inference: a sequence of measures converges just in case the measures assigned to each specific event converge. This does not generally hold in the case of weak convergence. For example, the Dirac distribution at $\frac{1}{n}$, $\delta_{\frac{1}{n}}$ converges weakly to δ_0 but does not converge setwise since $\delta_{\frac{1}{n}}(\{0\}) = 0$ for all n but $\delta_0(\{0\}) = 1$.

Example B.3. For example that $\Theta = [0, 1]$ on $\Omega = 2^\omega$. Let

$$E_s = \{x \in \Omega \mid s \text{ is a prefix of } x\}$$

and let $|s|_1$ = Number of 1’s in s and $|s|_0$ = Number of 0’s in s . Then $\mathbb{P}_\theta(E_s) = \theta^{|s|_1} \times (1 - \theta)^{|s|_0}$.

Then the topology (Θ, sw) is isomorphic to the Euclidean topology $(\Theta, |\cdot|)$. $\square$

Having the confidence property constrains the structure of c .

Theorem B.4. Suppose that Θ is topologized by setwise topology and has no isolated points. Let $c : \Sigma \rightarrow 2^\Theta$ have the confidence property at level α . Then for all θ ,

$$\mathbb{P}_\theta(c(E) \text{ has nonempty interior}) \geq 1 - \alpha.$$

Proof. Suppose for contradiction that there is a θ_0 such that $\mathbb{P}_{\theta_0}(c(E) \text{ has empty interior}) > \alpha$. By countable additivity there exist finitely many events $E_1, \dots, E_k \in \Sigma$ such that $c(E_i)$ has empty interior and $\mathbb{P}_{\theta_0}(\bigcup E_i) > \alpha$. Since Θ has no isolated points, there exists a neighborhood U of θ_0 containing a convergent sequence $\mu_k \rightarrow \theta_0$ with $\mathbb{P}_{\mu_k}(\bigcup E_i) > \alpha$ for all k . Since each $c(E_i)$ has nonempty interior, the relative complement $U \setminus \bigcup E_i$ is dense in U . Therefore there exists some $\mu \in U$ such that

- (1) $\mathbb{P}_\mu(\bigcup E_i) > \alpha$ and
- (2) $\mu \notin c(E_i)$ for any $i \leq k$.

By construction, the event $\mu \notin c(E)$ contains $\bigcup E_i$, so

$$(29) \quad \begin{aligned} \mathbb{P}_\mu(\mu \notin c(E)) &\geq \mathbb{P}_\mu(\bigcup E_i) \\ &> \alpha, \end{aligned}$$

violating the confidence property at level α . $\square$

As a corollary,

Corollary B.5. Let $c : \Sigma \rightarrow 2^\Theta$ have the confidence property at level α . Then the **topological confidence property at level 2α** holds: for all θ ,

$$\mathbb{P}_\theta(\theta \in c(E) \text{ and } c(E) \text{ has nonempty interior}) \geq 1 - 2\alpha.$$